

Computer Vision - Questions and Answers

Question: 1. A mobile app enhances photos automatically. Identify whether this task belongs to image processing or computer vision and justify.

Answer: Image Processing. Justification: The app enhances photos (brightness, contrast, filters, etc.) without understanding the image content.

Question: 2. A game engine generates a 3D virtual environment. Classify this under computer graphics or vision and explain.

Answer: Computer Graphics. Justification: It creates synthetic 3D environments rather than analyzing real-world images.

Question: 3. A system detects edges in an image before object recognition. Which level of vision is involved?

Answer: Low-level vision. Justification: Edge detection is a basic pixel-level operation.

Question: 4. A robot identifies objects after extracting features. Which vision level is this?

Answer: High-level vision. Justification: Object identification requires interpretation.

Question: 5. A CCTV system identifies a person from a crowd. Which vision level is involved?

Answer: High-level vision. Justification: Person recognition requires semantic understanding.

Question: 6. A camera captures a blurred image due to improper focus. What aspect of image formation is affected?

Answer: Focus (image sharpness). Justification: Incorrect lens focus causes blur.

Question: 7. A smartphone camera shows distorted images. What is the need for camera calibration here?

Answer: To correct lens distortion and improve image accuracy.

Question: 8. A self-driving car uses two cameras to estimate distance. What concept is being used?

Answer: Stereo vision. Justification: Two cameras estimate depth using disparity.

Question: 9. A drone captures images from different angles to build a 3D model. Which concept is applied?

Answer: Structure from Motion (SfM).

Question: 10. A medical imaging system analyzes MRI scans to detect tumors. Which CV application area does this belong to?

Answer: Medical Image Analysis.

Question: 11. A virtual try-on app overlays glasses on a user's face in real time. Which fields are involved: graphics, vision, or both?

Answer: Both Computer Vision and Computer Graphics.

Question: 12. A system converts a real-world 3D scene into a 2D image using a camera. What process of image formation is taking place?

Answer: Perspective projection.

Question: 13. A smartphone camera app corrects geometric distortion in captured images. Which concept is applied here?

Answer: Camera calibration (distortion correction).

Question: 14. A surveillance system detects motion but cannot identify objects. Which level of vision is represented?

Answer: Low-level vision.

Question: 15. A robot identifies objects after analyzing their shape and features. Which level of vision is involved?

Answer: High-level vision.

Question: 16. A VR system creates a virtual environment without using real-world images. Is this computer graphics or vision? Justify.

Answer: Computer Graphics. Justification: It creates synthetic scenes.

Question: 17. A stereo camera system generates depth maps for a scene. Which concept is being applied?

Answer: Stereo vision / Depth estimation.

Question: 18. A system analyzes multiple images of a building to estimate its structure. Which concept is used here?

Answer: Structure from Motion (SfM).

Question: 19. A biometric system uses facial features to verify identity. Which application domain of computer vision does this belong to?

Answer: Biometrics / Face Recognition.

Question: 20. A camera fails to measure accurate depth due to improper alignment of lenses. Which concept needs correction?

Answer: Stereo camera calibration.